

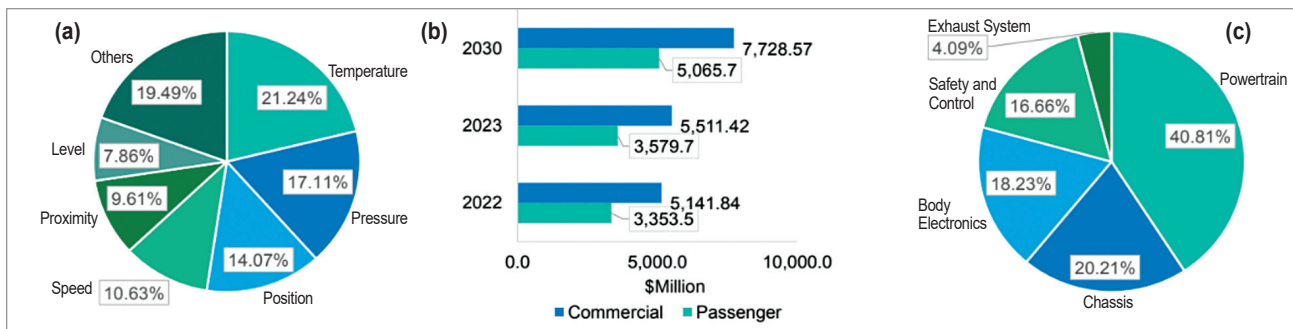


Fig. 3: (a) Global conventional sensors (by sensor type), value share, 2022 (% of global revenue), (b) Global conventional sensors (by vehicle type), and (c) Global conventional sensors (by application type), value share, 2022 (% of global revenue)

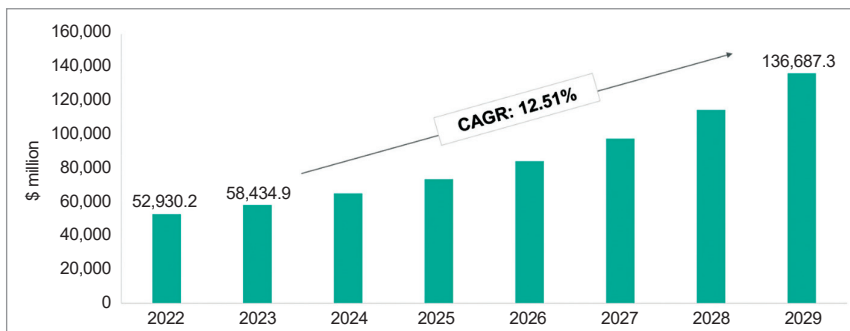


Fig. 4: Global advanced sensor market scenario—present and projected

is expected to increase, reaching \$136 billion by 2029, driven by the adoption of level two and level three autonomous vehicles.

Power electronics. Power electronics include onboard chargers, inverters, integrated antenna systems, and power management ICs. The power electronics market is set to grow at a CAGR of 14.18% over the next decade due to the increased adoption of electric, connected, and autonomous vehicles.

Within the sphere of conventional sensors, recent data paints a picture of a global market valued at approximately \$8.5 billion in 2022. As shown in Fig. 3, looking forward to 2030, this figure is poised to soar to an estimated \$13 billion.

While this growth trajectory certainly holds promise, it is essential to note that the compounded annual growth rate characterising the conventional sensors market presently maintains a conservative trajectory. A closer examination of

regional dynamics (shown in Fig. 2) reveals the Asia Pacific region, with a particular emphasis on China, as a prominent epicentre of automotive production. This region proudly claims the largest market share, propelled not only by the demand for luxury and premium vehicles but also by the sheer volume of vehicles being adopted across Asian markets. Within the categorisation of conventional sensors, temperature sensors, pressure sensors, and position sensors emerge as the primary driving forces, collectively sculpting the landscape of this pivotal sector within the industry.

In advanced sensors, numerous drivers, restraints, and opportunities are at play, including some that require closer examination. Notably, a persistent challenge is the high cost associated with advanced sensors. In the past, lidar sensors commanded prices exceeding \$5000 to \$10,000. However, recent advancements have seen prices drop

significantly, with some companies offering lidar sensors for around \$1800 or less. This cost sensitivity poses a dilemma for manufacturers, as automotive OEMs are hesitant to allocate substantial budgets per vehicle for these advanced sensors. As a result, manufacturers of lidar and advanced sensors have prioritised cost optimisation and production efficiency.

Moreover, an emerging opportunity lies in the utilisation of 5G technology and the impending 6G developments, which are poised to revolutionise connectivity and facilitate seamless cloud synchronisation. This technological advancement promises to enhance the performance of advanced sensors, enabling real-time data processing with precision. These pivotal insights surfaced during our discussions with advanced sensor manufacturers.

The market dynamics in 2022 and the preceding year have shown the advanced sensors market was valued at approximately \$50 to \$52 billion.

Fig. 4 shows the global advanced sensor market scenario—present and projected. Projections point to an impressive surge, with the market expected to reach a staggering \$136 billion by 2029. It is worth noting that this growth trajectory outpaces that of conventional sensors, with the compounded annual growth rate (CAGR) for advanced